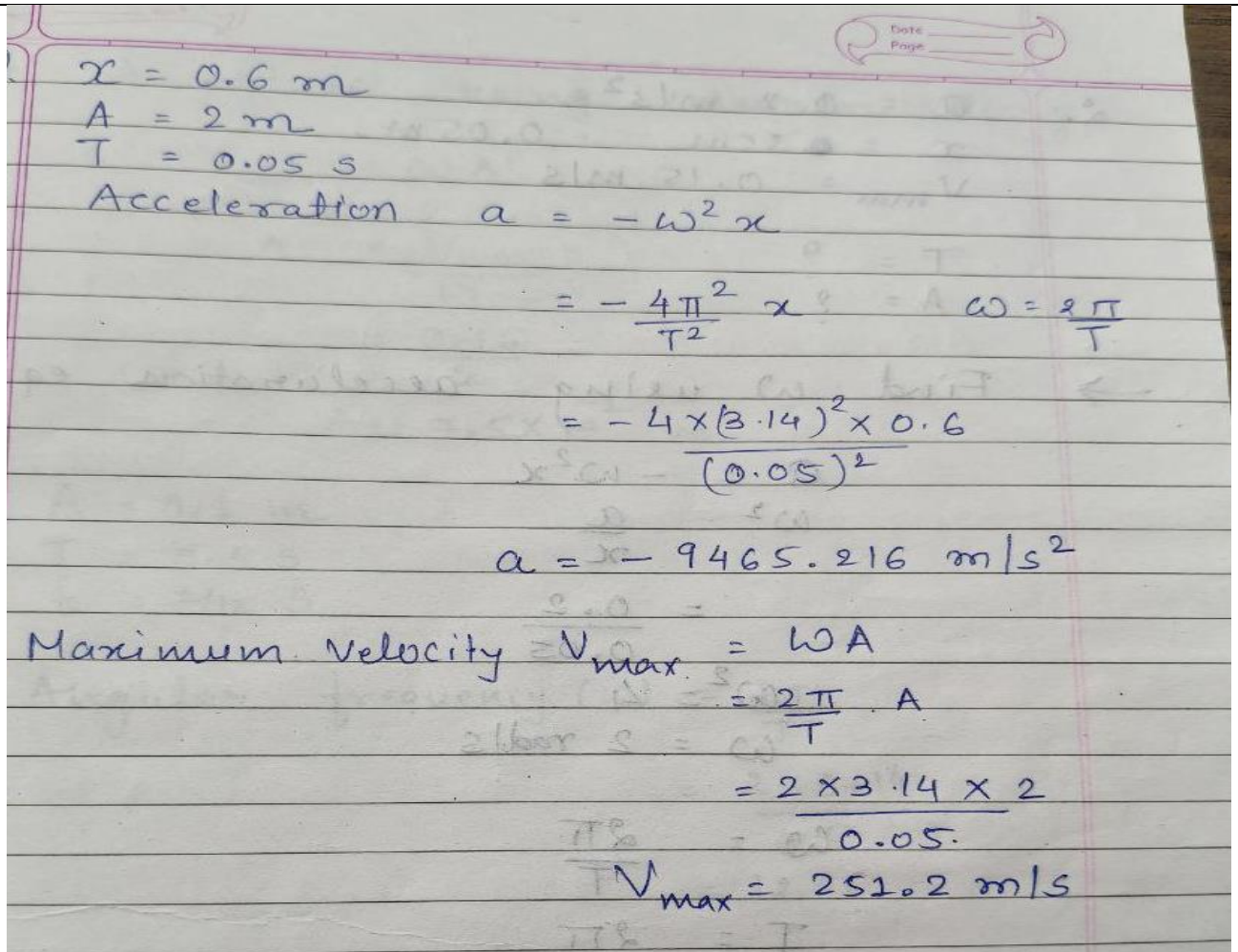


Examples (WAVES AND MOTION)

1. Find the acceleration of particle performing simple harmonic motion (SHM) when it is at 0.6 m from its mean position. The time period of SHM is 0.05 sec. Also calculate maximum velocity if the amplitude of SHM is 2 m.

(Jun-2019)(Aug-2023)[NLJIET]

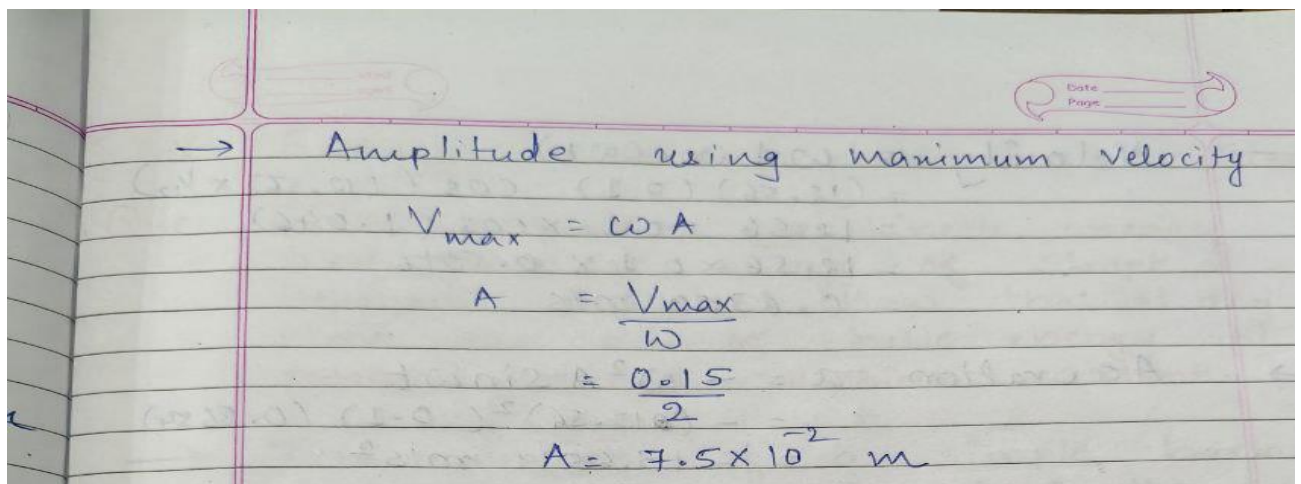


Handwritten solution for the first part of the problem:

$$\begin{aligned}
 x &= 0.6 \text{ m} \\
 A &= 2 \text{ m} \\
 T &= 0.05 \text{ s} \\
 \text{Acceleration } a &= -\omega^2 x \\
 &= -\frac{4\pi^2}{T^2} x \quad \omega = \frac{2\pi}{T} \\
 &= -\frac{4 \times (3.14)^2 \times 0.6}{(0.05)^2} \\
 a &= -9465.216 \text{ m/s}^2
 \end{aligned}$$

Maximum Velocity $V_{\max} = \omega A$

$$\begin{aligned}
 &= \frac{2\pi}{T} \cdot A \\
 &= \frac{2 \times 3.14 \times 2}{0.05} \\
 V_{\max} &= 251.2 \text{ m/s}
 \end{aligned}$$



Handwritten solution for the second part of the problem:

→ Amplitude using maximum velocity

$$\begin{aligned}
 V_{\max} &= \omega A \\
 A &= \frac{V_{\max}}{\omega} \\
 &= \frac{0.15}{2} \\
 A &= 7.5 \times 10^{-2} \text{ m}
 \end{aligned}$$

2. A particle execute simple harmonic motion with amplitude of 0.1 m and a period of 0.5 s. Calculate its displacement, velocity and acceleration (1/12) s after it crosses the mean position. (Jan-2024) [NLJIET] 3

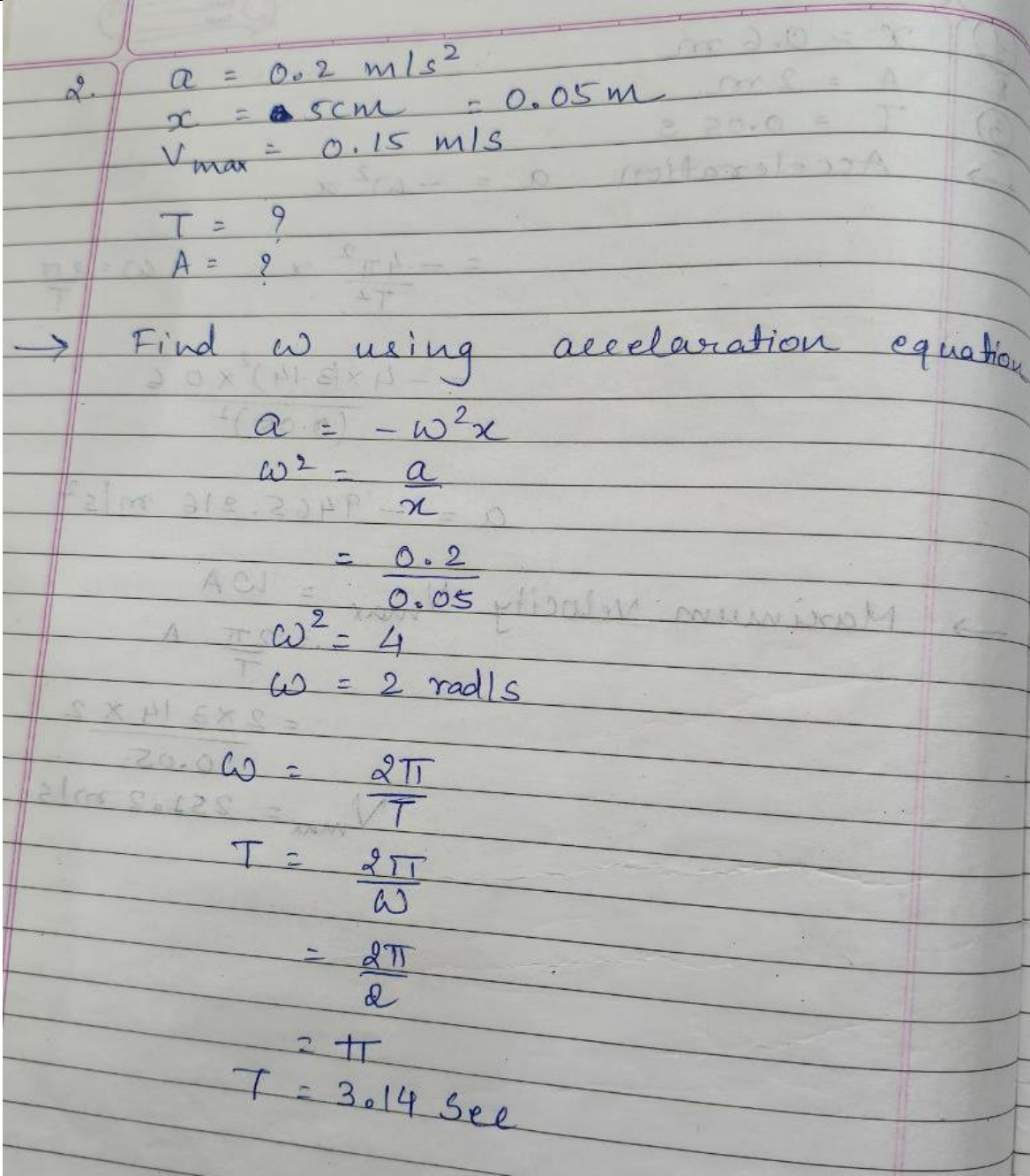
4. $A = 0.1 \text{ m}$
 $T = 0.5 \text{ s}$
 $t = 1/12 \text{ s}$

→ Angular frequency (ω) = $\frac{2\pi}{T}$
 $= \frac{2 \times 3.14}{0.5}$
 $= 12.56$

→ Displacement $x = A \sin \omega t$
 $= 0.1 \sin (12.56 \times 1/12)$
 $= 0.1 \sin (1.046)$
 $= 0.1 (0.8654)$
 $= 0.08654 \text{ m}$

→ Velocity = $\omega A \cos(\omega t)$
 $= (12.56) (0.1) \cos (12.56 \times 1/12)$
 $= 12.56 \times 0.1 \times \cos (1.046)$
 $= 12.56 \times 0.1 \times 0.5016$
 $= 0.6300 \text{ m/s}$

→ Acceleration $a = -\omega^2 A \sin \omega t$
 $= -(12.56)^2 (0.1) (0.8654)$
 $= -13.6519 \text{ m/s}^2$

3.	An oscillator is performing SHM with an acceleration of 0.2 m/s^2 when displacement is 5 cm. Its maximum speed is 0.15 m/s. Find time period and amplitude of oscillator. (Aug-2022) [NLJIET]	4
	 $a = 0.2 \text{ m/s}^2$ $x = 5 \text{ cm} = 0.05 \text{ m}$ $V_{\max} = 0.15 \text{ m/s}$ $T = ?$ $A = ?$ → Find ω using acceleration equation $a = -\omega^2 x$ $\omega^2 = \frac{a}{x}$ $= \frac{0.2}{0.05}$ $\omega^2 = 4$ $\omega = 2 \text{ rad/s}$ $\omega = \frac{2\pi}{T}$ $T = \frac{2\pi}{\omega}$ $= \frac{2\pi}{2}$ $= \pi$ $T = 3.14 \text{ sec}$	
4.	A particle of mass 0.50 kg executes a simple harmonic motion. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of the motion. (force constant $k = 50 \text{ N/m}$). (Jun-2025-New) [NLJIET]	4

Here, $m = 0.5 \text{ kg}$ force constant $k = 50 \text{ N/m}$ amplitude of

$$v_{\max} = 10 \text{ m/s}$$

$$k = 50 \text{ N/m}$$

$$A = ?$$

$$v = \omega_0 \sqrt{A^2 - y^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}} = 10 \text{ rad/s} \quad (2 \text{ marks})$$

To obtain the amplitude

$$v_{\max} = \omega_0 A$$

$$A = \frac{v_{\max}}{\omega_0}$$

$$A = 1 \text{ m}$$

(2 marks)